

Muhammad Hammad Bashir

Machine Learning Engineer — Computer Vision — Generative AI

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with 5+ years of experience across Computer Vision, deep learning, and AI systems spanning sports analytics, Ads and marketing, Chatbots, agriculture, and multimodal AI. Specialized in end-to-end video intelligence pipelines involving object detection, tracking, pose estimation, and event recognition using YOLO, MediaPipe, ByteTrack, and deep learning frameworks. Proven track record in building scalable ML systems, real-time inference pipelines, multi-camera streaming analytics, and production-grade MLOps deployments.

PROFESSIONAL EXPERIENCE

Sportunity / Sports Analytics Platform

Computer Vision Engineer (Sports AI)

June 2025 – Present

- Built a real-time sports computer vision system for soccer, tennis, and volleyball using live RTSP (HIKvision) camera feeds.
- Implemented **YOLOv8m + ByteTrack + TrackNet** for player, ball detection, and multi-object tracking under low-latency constraints.
- Developed event detection pipelines including ball bounce detection, trajectory interpolation, speed/acceleration estimation, and shot classification.
- Designed multi-camera synchronization system for cross-angle tracking, enabling accurate match reconstruction and highlight generation.
- Integrated **MediaPipe** pose estimation for biomechanics analysis (jump, sprint, movement efficiency).
- Optimized inference pipeline using **ONNX + TensorRT** for edge and GPU deployment.
- Built automated AI coaching system generating real-time analytics dashboards for coaches and athletes.

Markify / CreativeScore AI

London, UK

AI Consultant (Computer Vision & Generative AI)

June 2023 – June 2025

- Developed neuromarketing AI systems analyzing ad effectiveness using computer vision and multi-modal signals (image, video, audio, and text).
- Built attention prediction and saliency mapping models to identify high-impact visual regions in advertisements.
- Designed emotion recognition pipelines for viewer reaction analysis using deep CNNs and transformer-based encoders.
- Implemented **Pix2PixHD**, GAN-based augmentation, and diffusion-based creative optimization for ad generation.
- Built end-to-end **FastAPI + AWS + Azure** CI/CD pipelines for scalable deployment.
- Delivered creative intelligence platform (creativescore.ai) for optimizing marketing performance and engagement.

LUMS / AI & Big Data Centre

Pakistan

Research Assistant (AI & Computer Vision)

Sept 2021 – June 2023

- Developed ML-based livestock health monitoring systems using image classification and sensor data.
- Worked on flood and drought prediction using **LSTM** and **RNN**-based time-series models.
- Built agricultural AI systems for crop monitoring and yield prediction using satellite imagery.
- Implemented classical ML pipelines using **Scikit-learn**, XGBoost, Flask, and SQL-based deployments.

Freelance ML Engineer

Upwork & Fiverr

Computer Vision & ML Consultant

Aug 2021 – Jul 2023

- Delivered end-to-end ML and computer vision solutions for global clients in sports analytics, document AI, and geospatial intelligence.

SKILLS

Computer Vision:	YOLOv8, OpenCV, MediaPipe, ByteTrack, DeepSORT, TrackNet, SAM, CLIP
Video Analytics:	RTSP pipelines, multi-camera sync, event detection, tracking systems
Deep Learning:	PyTorch, TensorFlow, CNNs, Transformers, GANs, diffusion models
MLOps:	Docker, CI/CD, Azure DevOps, MLflow, DVC
Deployment:	FastAPI, AWS, Azure, ONNX, TensorRT, serverless GPU systems
Geospatial AI:	QGIS, GDAL, rasterio, satellite image analysis
LLMs & Multimodal AI:	GPT, Claude, LLaMA, LangChain, CrewAI

EDUCATION

Namal University

BSc in Computer Science [Specialisation in Machine Learning]; CGPA: 3.46/4.0

Mianwali, Pakistan

2017 – 2021

SELECTED PROJECTS

Real-Time Multi-Camera Sports Vision System:

- Built a production-grade sports AI system using live RTSP feeds for real-time match analysis.
- Integrated **YOLOv8**, **ByteTrack**, and **TrackNet** for detection and tracking of players and ball.
- Developed ball interpolation, bounce detection, and event recognition modules.
- Used **MediaPipe** for pose-based biomechanics; enabled multi-camera synchronization for highlight generation and AI-based coaching feedback.

MoversAI – Intelligent Inventory Quote Generation System:

- Developed an AI-powered web platform allowing users to capture room-by-room images for automated moving quotes.
- Used object detection and scene understanding to identify items, estimate volume, and generate pricing.
- Integrated human-in-the-loop verification for accuracy and production reliability.

Drone & Satellite Vision Intelligence (TMGDronity):

- Worked on multimodal geospatial AI system combining satellite, drone, RGB, multispectral, and hyperspectral imagery.
- Implemented geospatial alignment using **QGIS**, **GDAL**, and **rasterio**.
- Built **YOLO**-based object detection and **UNet** segmentation models for vegetation, infrastructure, and underground object inference.
- Included greenery estimation and plant counting pipelines.

DoctorAI – Fall Detection & Health Monitoring System:

- Developed AI-assisted healthcare monitoring system combining computer vision and LLM-based reasoning.
- Implemented fall detection using pose estimation and anomaly detection pipelines.
- Integrated multimodal signals for patient activity monitoring and alert generation.

CERTIFICATIONS

- Completed the **AWS Machine Learning Foundations** course in the Udacity Scholarship Program.
- Earned the **Deep Learning Specialization** certification from **DeepLearning.AI** (Coursera).
- Completed the **Machine Learning Engineering for Production (MLOps)** Specialization by **DeepLearning.AI** on Coursera.
- Completed the **Machine Learning Specialization** by **DeepLearning.AI** and Stanford Online (Coursera).
- Completed the **Computer Vision** course on Coursera.